

# Future State Maintenance (FSM) PoC

Analytics & Monitoring — Demo Screens Specification

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This document specifies the screens the development team should build for the client demo of the Analytics & Monitoring capability. It covers the end-to-end monitoring workflow the screens need to support, then a detailed spec for each screen — what's shown, what drives it, and how it should behave.

Reference client: Digital Realty · Site scope: EMEA, NAM, APAC · Prepared for: Development team

# 1. End-to-end monitoring workflow

Every screen in this spec is a window into one continuous pipeline. Build the screens so they can all be driven from the same underlying data model — an asset moves through these stages, and each screen just surfaces a different slice of it.

## 1.1 Detection phase

| Step                     | What happens                                                                                                                                            |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuous monitoring    | Vibration, thermal, and ultrasound sensors stream data continuously for each asset.                                                                     |
| Anomaly detected         | One or more sensor readings trend outside baseline (e.g. vibration amplitude rising over multiple days).                                                |
| Cross-sensor correlation | The other two sensor types are checked against the same asset over the same window. Agreement across sensors raises confidence; disagreement lowers it. |
| Failure code assigned    | A standardized failure code is matched from the pattern (e.g. VIB-BRG-04 = bearing wear) with a confidence percentage.                                  |
| Health score updated     | The asset's health score drops in proportion to failure code severity and confidence.                                                                   |

## 1.2 Response and feedback phase

| Step                      | What happens                                                                                                        |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| Alert generated           | Pushed to the dashboard and stakeholder feed once confidence and severity cross a threshold.                        |
| Engineering review        | An engineer checks the alert against trend data and either validates or dismisses it.                               |
| Recommendation issued     | A specific, actionable maintenance recommendation is generated (what to check, how urgent).                         |
| Work order and inspection | A work order is dispatched; a technician inspects the asset and logs a finding.                                     |
| Outcome logged            | The finding (confirmed / false positive / different issue) is written back and used to tune thresholds and scoring. |

## 2. Demo screens

Six screens cover the full workflow above. Build them to share one data model so navigating between them (e.g. clicking an alert to see the asset it belongs to) is a filter change, not a different system.

### 2.1. Asset overview / fleet view

**Purpose:** The landing screen for the demo. Shows every monitored asset at a glance so the client immediately sees breadth of coverage before drilling into any one asset.

| UI element           | Description                                             | Behavior / status logic                                                                                                              |
|----------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Asset list/grid      | One row or card per asset: ID, name, site, asset class. | Sortable by health score, filterable by site/class/status.                                                                           |
| Health score badge   | Current score (0-100) with color band.                  | Green $\geq 85$ , amber 60-84, red $< 60$ .                                                                                          |
| Status indicator     | Healthy / warning / critical / alert active.            | Derived from latest alert severity, not just health score, so a fresh low-severity alert still shows distinctly from a critical one. |
| Last alert timestamp | When the asset last triggered an alert, if any.         | Blank/'None' state for assets with a clean history.                                                                                  |
| Site/region filter   | Filter by EMEA/NAM/APAC and by data hall.               | Supports the multi-site PoC scope.                                                                                                   |

**Data source:** Assets table + latest row from Health\_Score\_History + latest row from Alerts, joined on Asset\_ID.

**Dev notes:** This is the screen the client sees first — keep it fast and uncluttered. Clicking a row routes to 2.2 for that asset.

### 2.2. Asset detail / live monitoring

**Purpose:** The deep-dive screen for one asset. This is where the demo spends the most time — it should tell the full current story of the asset in one view.

| UI element           | Description                                                                             | Behavior / status logic                                                                                                                            |
|----------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Asset header         | Name, ID, asset class, site, criticality.                                               | Static from asset master data.                                                                                                                     |
| Health score panel   | Current score plus trend arrow (rising/declining/stable).                               | Trend = comparison to score 7 days prior.                                                                                                          |
| Sensor readout cards | One card per sensor: vibration (mm/s), thermal ( $^{\circ}\text{C}$ ), ultrasound (dB). | Each card shows current value + status (normal/elevated/above threshold) using per-asset baseline thresholds, not one fixed number for all assets. |
| Active alert banner  | Shown only if an open alert exists: failure code, description, confidence.              | Hidden entirely for healthy assets — do not show an empty/placeholder alert box.                                                                   |
| Recommendation card  | Suggested action tied to the active alert, with an urgency label.                       | Only rendered alongside an active alert.                                                                                                           |

**Data source:** Assets + latest Sensor\_Readings row + latest Health\_Score\_History row + open row from Alerts + matching row from Recommendations\_WorkOrders.

**Dev notes:** Build the three sensor cards as one reusable component parameterized by sensor type — this same component is what powers the trend screen (2.3) when expanded.

## 2.3. Trend analysis

**Purpose:** Time-series view proving early detection — this is the screen that shows the client we caught something *\*before\** it became obvious, not just that we detected it eventually.

| UI element         | Description                                                                | Behavior / status logic                                                                 |
|--------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Sensor selector    | Toggle between vibration / thermal / ultrasound for the selected asset.    | Default to whichever sensor triggered the most recent alert.                            |
| Time-series chart  | Daily readings over the monitoring window (at least 30 days for the demo). | Overlay a baseline band and a threshold line so the crossing point is visually obvious. |
| Annotation marker  | Point on the chart marking when the alert fired.                           | Pulled from Alerts.Date_Raised for that asset/sensor.                                   |
| Date range control | Zoom/filter the window shown.                                              | Not critical for demo but expected by reviewers.                                        |

**Data source:** Sensor\_Readings filtered by Asset\_ID and sensor column, plus Alerts.Date\_Raised for the marker.

**Dev notes:** This is the single most persuasive screen in the demo — prioritize it if the team is short on time. Even a static rendering of the CRAH-01 vibration trend with the threshold crossing marked is worth more than a polished but empty fleet view.

## 2.4. Alert and failure code detail

**Purpose:** Explains exactly what was detected and why the system believes it, in language an engineer would trust.

| UI element           | Description                                      | Behavior / status logic                                                                                               |
|----------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Failure code         | Standardized code, e.g. VIB-BRG-04.              | Pulled from a fixed reference list per asset class/sensor (see Section 4.2-4.4 of the SOW for the failure mode list). |
| Triggering sensor(s) | Which sensor(s) contributed to this alert.       | Show all sensors that agreed during cross-sensor correlation, not just the first one to trip.                         |
| Confidence score     | Percentage confidence in the failure code match. | Should visibly increase when a second/third sensor corroborates the first (e.g. 82% → 91%).                           |
| Severity             | Critical / warning / informational.              | Drives the color used on the fleet view badge.                                                                        |
| Status               | Open / under review / dispatched / closed.       | Changes as engineering review and the work order progress.                                                            |

**Data source:** Alerts table, joined to a static Failure\_Code reference table for the human-readable description.

**Dev notes:** Keep the confidence-boost behavior visible somewhere in the UI (even a small 'confirmed by 2 of 3 sensors' label) — it's the detail that sells the cross-sensor correlation story.

## 2.5. Maintenance recommendation / work order

**Purpose:** Shows the alert turning into an actual maintenance action — this is what proves the system is useful to the maintenance team, not just informative.

| UI element         | Description                                                             | Behavior / status logic                                                                           |
|--------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Recommended action | Plain-language suggested action, e.g. 'Inspect bearing within 2 weeks'. | One recommendation per alert; do not stack multiple vague suggestions.                            |
| Urgency            | High / medium / low.                                                    | Derived from severity + confidence, not set independently.                                        |
| Work order status  | Open / dispatched / completed.                                          | Updates as the technician workflow progresses; demo should show at least one asset at each stage. |
| Technician finding | Free-text finding logged after inspection.                              | Only populated once status = completed.                                                           |
| Outcome            | True positive / false positive / different issue found.                 | This is the field that feeds Section 2.6.                                                         |

**Data source:** Recommendations\_WorkOrders table, joined to Alerts for context.

**Dev notes:** Show three different states across the demo assets (open, dispatched, completed) rather than three identical 'completed' rows — it's more convincing that the workflow is real.

## 2.6. Correlation / outcome log

**Purpose:** The accuracy proof screen. This is what the client's Go/No-Go decision leans on, so it should read as a track record, not a single anecdote.

| UI element                   | Description                                                                  | Behavior / status logic                                                                       |
|------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Log table                    | One row per closed alert: asset, failure code, predicted vs. actual finding. | Sort by date, most recent first.                                                              |
| Accuracy summary             | Rolling alert accuracy % and false positive rate %.                          | Compute against the KPI targets: accuracy 70-80%, false positive rate 10-20% (see Section 3). |
| Filter by asset class / site | Slice the log by mechanical vs. electrical, or by site.                      | Optional for demo, useful if the client asks a follow-up question live.                       |

**Data source:** Recommendations\_WorkOrders where Outcome is populated, aggregated for the summary metrics.

**Dev notes:** Even with only 2-3 closed records in the demo data, present this as a running log, not a one-off case study — frame the UI as something that accumulates over the 12-month PoC.

### 3. KPI reference for these screens

Every screen above should be able to report against these targets from the SOW. Build the underlying metrics (accuracy, false positive rate, availability) as queries the whole app can reuse, rather than one-off calculations per screen.

| KPI                          | Target   | Primary screen(s) |
|------------------------------|----------|-------------------|
| Sensor availability          | ~90-100% | 2.1, 2.2          |
| Data acquisition reliability | ~90-100% | 2.1, 2.2          |
| Alert accuracy               | ~70-80%  | 2.6               |
| False positive rate          | ~10-20%  | 2.6               |
| Dashboard availability       | ~90-100% | All screens       |